open forcefield

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Consortium Roadmap 2026-2027

Initial Discussion

Today's agenda

What we've prepared

- Overview of the roadmap process
- 10 minute presentation of staff ideas for next year's priorities
  - Recommendations (highlighted) and alternatives
- Open discussion

What we need from you

- Questions and comments about scope of proposed items
- Ideas for objectives / deliverables we haven't thought of
- Identification of "do not do"s
- Feedback on priorities

OpenFF Roadmap

Long-term roadmap is consistent

Expanding Domain of Applicability

Covering more classes of biomolecule

- Peptides and proteins
- Lipids
- Nucleic Acids

Improving Accuracy

Identifying and correcting errors

Developing new fitting targets

Enhancing methods

Enhancing Usability

Lowering barriers to use

Roadmap Process

Roadmap is prioritized annually

- Staff brainstorming
- Lead team filtering
- Advisory Board feedback
- Lead team proposal
- Governing Board vote

Baseline Effort

Basic Functions of the Consortium

Incremental improvements to existing flagship force field

Ongoing maintenance of software stack

Facilitating science projects in the OpenFF Initiative

Streamline Processes

Finish building smee+descent based fitting stack

Deliverable: Replace all use of ForceBalance in our main-line force field fits with smee and descent, creating functionality for condensed phase fitting.

Replace NAGL with either next-gen infra or minimal implementation

Deliverable: Either refactor out unmaintained deps (DGL), or collaborate with academics to make a next-gen replacement with maintained deps.

Refactor and simplify QCSubmit

Deliverable: Remove unused functionality and refactor to better match design patterns in modern QC ecosystem

Refactor OpenFF Toolkit

Deliverable: Consolidate interchange into openff-toolkit. Go RDKit-only (keep OE wrapper in other repo) to encourage external contributors.

Implementing surrogate modelling

Deliverable: a prototype workflow and software using surrogate modelling in place of more expensive targets

Expand Domain of Applicability

A peptide forcefield

Deliverable: benchmark SMIRNOFF candidates on peptides and release if performance is good

Lipid parameters

Deliverable: supporting academic collaborators in developing, benchmarking, and releasing a co-optimized small molecule + membrane force field

A folded protein force field

Deliverable: supporting academic collaborators in developing, benchmarking, and releasing a co-optimized small molecule + folded protein force field

Nucleic acid QM datasets

Deliverable: dataset planning and creation

Improve Accuracy

Improve parameters for alcohols and amines

Deliverable: a released force field with improved performance on alcohol and amine benchmarks

Co-optimized water model

Deliverable: a 3-site co-optimized water model with improved performance on benchmarks

Next-generation charge model fit to QM

Deliverable: neural network charges on modern GNN packages, fit directly to QM properties

Adopt Presto ("BespokeFit 2")

Deliverable: Adopt "Presto" (Cole lab's successor to Bespokefit). Uses high temperature MD instead of torsion drives, and fast ML potentials.

Revisit continuous typing with Espaloma

Deliverable: supporting academic collaborators in further experiments with Espaloma

Co-optimized ion parameters

Deliverable: supporting academic collaborators in re-fitting ion parameters to new targets

Virtual sites

Deliverable: a virtual site force field with virtual sites addressing sigma holes and N lone pairs

Wide-ranging internal benchmarking

Deliverable: re-running benchmarks of internal (Parsley+Sage) and external (eg GAFF and CGENFF) against current benchmarks (QM, condensed phase, HFEs, SFEs, and RBFEs)

Industry benchmarking collaboration

Deliverable: a public benchmarking effort with partner companies on QM molecule energetics (other recommendations?)

Data-driven SMIRKS typing

Deliverable: a prototype force field where SMIRKS are automatically generated from data

Investigating different mixing rules

Deliverable: a prototype force field trained with different mixing rules

Enhance User Experience

PyPI packaging

Deliverable: Move core packages to PyPI (+/- AmberTools)

User group meeting

Deliverable: Host a multi-day in person event focused on users and developers, and invite other OMSF projects to participate.

Improve training materials

Deliverable: Refactor documentation and examples, convert notebooks to Marimo

Staff Recommendations

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